Wisair 502 Integrated RF Transceiver

Pre-production Low Power MBOA UWB RF Chip

General Description

Wisair 502 is a Multi-Band OFDM monolithic RF transceiver chip, part of Wisair's Ultra Wideband (UWB) chipset solution.

Integrated into Wisair's Development Kit (DVK) and reference designs, the Wisair 502 is WiMedia\MBOA compliant, enabling the rapid development of UWB-based consumer electronics and wireless USB applications running on mobile devices, laptops, PC's, and PC peripherals.

The 502 chip is a fully monolithic transceiver implemented on a SiGe-BiCMOS process, designed for reduced power consumption and low-cost UWB solutions. It supports transmitted data rates up to 480Mbps using MB-OFDM TFI and FFI modes and occupies a frequency spectrum range between 3.1GHz to 4.8GHz, with three sub-bands of 528 MHz each.

The RF chip incorporates an on-chip RF band pass filter (BPF), a broad-band Receiver with a wide programmable dynamic range, an ultra-fast hopping broadband Synthesizer, including an on-chip VCO, and a programmable Power Amplifier (PA), to ensure maximum authorized output power under all conditions.

The external supporting components are passive. The chip also uses a single off-the-shelf crystal. The transceiver chip can be customized to meet different regional regulations and certifications requirements worldwide, through a pre-selected or programmable option.

The 502 chip incorporates internal filtering providing co-existence with Bluetooth as well as 802.11a/b/g.

Key Features

- Pre-production integrated RF IC
- Compliant with MBOA PHY spec. Rev. 1.0
- Low current consumption
- Selectable multiple rates, up to 480Mbps
- 3.1GHz to 4.8GHz frequency range supporting 3 sub-bands, 528MHz each
- Supports TFI and FFI modes
- Support for co-existence with Bluetooth and 802.11a/b/g
- Integrated transceiver with on-chip VCO and internal BPF
- Single off-the-shelf crystal
- Differential antenna output
- Antenna diversity
- Does not require external BPF or Balun for most applications
- Low bill-of-material <10 components
- Based on IBM 0.18um SiGe RF technology
- 48 Leadless in QFN package

Applications

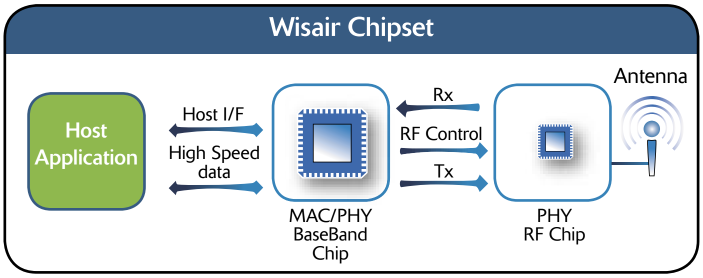
- **Home & Office Wireless USB (WUSB)** Supports High, Full & Low speeds (480 Mbps, 12 Mbps, 1.5 Mbps)
- **Home wireless video and high quality audio** Multiple HDTV streams with full quality of service (QOS)
- **Fast download data transfer** Synchronization with multiple devices in the home/ office environment, supporting battery powered mobile devices including; Cellular phones, PDA's and A/V handheld players.
- **Home wireless Multi-Streaming** Concurrent synchronized multiple broadband multi-media streams
- **Wireless ad-hoc gaming networks** for portable gaming consoles

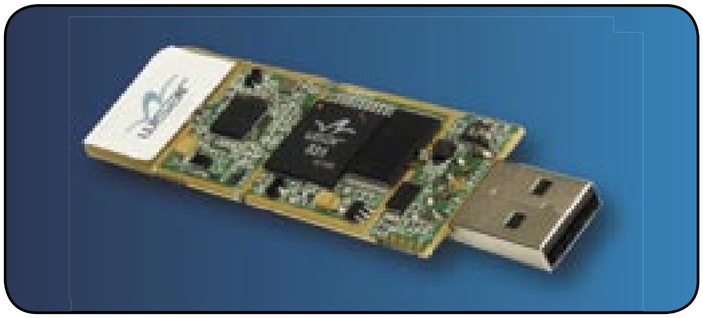
Wisair Multiband OFDM UWB Chipset

Wisair offers a comprehensive solution based on a UWB chipset, software API, and a reference design system for demonstration, evaluation and product development purposes.

The Wisair UWB chipset provides a high bit rate, low power consumption and low cost solution for Wireless Personal Area Network (WPAN) applications. It supports short-range home/office wireless connectivity, multi-streaming of high quality video, fast upload/download of content and broadband wireless multi-media sharing.

Wisair's chipset consists of:
- Wisair 502 PHY RF chip
- Wisair 531 MAC\ Baseband chipset





480 Mbps Ultra Wideband USB host dongle reference design

Reference Design

Wisair reference designs target wireless USB applications and fast Ethernet/IP communication. Using the Wisair chipset in an optimal way to minimize the Bill-of-Material (BOM), size and power consumption, reference designs include the 531 MAC\Baseband chip, the 502 transceiver IC and a differential antenna connector. Different antennas are available as part of the reference design.

External supporting components include a single off-the-shelf crystal and passives. The assembled 531 and 502 reference design circuitry covers a small footprint on the board's space of 21 by 32 mm. Also included is a software API and WisMan™ Windows-based reference application for controlling and managing the chipset.

About Wisair

Wisair develops and markets Ultra Wideband, high performance wireless chipset solutions. The company is an active promoter member of the WiMedia Alliance, an active participant in the IEEE standardization process, and a member of the Wireless USB promoter group.

A privately-held company, and part of the RAD group, Wisair is headquartered in Israel with offices in USA, Japan, Korea and Taiwan.